

Velvet examples book

Generated from openfluke/example - every chapter README plus live go run output.

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cam/02_combine - Combine strategies [PASS]
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cam/04_kit - CamKit [PASS]
cam/05_layers - Layers as cams [PASS]
cam/06_recipes - Recipes [PASS]

1. What Velvet is

suite=welvet slug=01-welvet status=PASS time=0.08s

README

1. What Velvet is

Part: I - Orientation

Package: github.com/openfluke/welvet

Status: ok - [OK] engine

When

You need github.com/openfluke/welvet.

Where

import "github.com/openfluke/welvet/github.com/openfluke/welvet"

```
cd 01-welvet && source ../env.sh && go run .
```

Why

Loom's flat poly/ package hit import-cycle and honesty walls (QAT morph, silent fallbacks, god-layer). Velvet is the rewrite: one feature per folder, storage-truth dtypes/quants, Dense as the shared MatVec microkernel, tests only in w2a, apps only in apps/.

What

An AI engine in Go: layers, 34 dtypes, 20 quant formats, and three backends (CPU tiled - Plan 9 SIMD - WebGPU). Version tracks a 100-point scorecard (today v1.1.0 - 100/100).

Sample output (captured)

```
pre 4 post 4
```

Live capture from go run ./cmd/runall on local ../../welvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/01-welvet).

Live output

```
pre 4 post 4
```

2. Repository map

suite=welvet slug=02-tree status=PASS time=0.04s

README

2. Repository map

Part: I - Orientation

Package: -

Status: ok - [OK] layout

When

Orientation chapter: Repository map.

Where

Book / repo map (no single import)

```
cd 02-tree && source ../env.sh && go run .
```

Why

Readers need a single map of what is engine vs harness vs app vs stub.

What

Top-level folders and their ownership. Engine packages never import w2a.

Sample output (captured)

```
import github.com/openfluke/welvet/core/...
import github.com/openfluke/welvet/weights/...
import github.com/openfluke/welvet/quant/...
import github.com/openfluke/welvet/simd/...
import github.com/openfluke/welvet/webgpu/...
import github.com/openfluke/welvet/tiling/...
import github.com/openfluke/welvet/architecture/...
import github.com/openfluke/welvet/fusedgpu/...
import github.com/openfluke/welvet/layers/...
import github.com/openfluke/welvet/runtime/...
import github.com/openfluke/welvet/systems/...
import github.com/openfluke/welvet/lucy/...
import github.com/openfluke/welvet/model/...
```

Live capture from go run ./cmd/runall on local ../.. /welvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/02-tree).

Live output

```
import github.com/openfluke/welvet/core/...
import github.com/openfluke/welvet/weights/...
import github.com/openfluke/welvet/quant/...
import github.com/openfluke/welvet/simd/...
import github.com/openfluke/welvet/webgpu/...
import github.com/openfluke/welvet/tiling/...
```

```
import github.com/openfluke/welvet/architecture/...
import github.com/openfluke/welvet/fusedgpu/...
import github.com/openfluke/welvet/layers/...
import github.com/openfluke/welvet/runtime/...
import github.com/openfluke/welvet/systems/...
import github.com/openfluke/welvet/lucy/...
import github.com/openfluke/welvet/model/...
```

3. core — types & backends

suite=velvet slug=03-core status=PASS time=0.04s

README

3. core - types & backends

Part: II - Foundation

Package: github.com/openfluke/velvet/core

Status: ok - [OK]

When

You need the core foundation package.

Where

```
import "github.com/openfluke/velvet/core"
```

```
cd 03-core && source ../env.sh && go run .
```

Why

Every polymorphic path needs one place for DType, LayerType, Activation, Backend, Tensor[T], and slim Layer metadata - without QAT morph defaults.

What

34 storage dtypes, Numeric generics, Tensor[T], ExecConfig (BackendCPUTiled | BackendSIMD | BackendWebGPU), Activate/ActivateDeriv, and converters for f16/bf16/fp8/fp4.

Sample output (captured)

```
simd bfloat16 8
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/03-core).

Live output

```
simd bfloat16 8
```

4. weights — FormatNone MatVec

suite=welvet slug=04-weights status=PASS time=0.04s

README

4. weights - FormatNone MatVec

Part: II - Foundation

Package: github.com/openfluke/welvet/weights

Status: ok - [OK]

When

You need the weights foundation package.

Where

```
import "github.com/openfluke/welvet/weights"
```

```
cd 04-weights && source ../env.sh && go run .
```

Why

Unquantized matrices still need a typed store that streams MatVec and SGD without forcing a float32 master or Morph-as-training.

What

Store holds DType + Format + Native/Packed bytes. New[T], MatVec, MatVecT, DecodeRow, SelectWire (F32/F64/I8), ApplySGD. FormatNone: update in native lanes (float32 payload is the only f32 buffer; float64 uses native ALU; other dtypes decode->update->re-encode). Packed: unpack->update->re-Pack then drop scratch. RetainsF32Master() is true only for FormatNone+float32. Dense and composite projs share this store.

Sample output (captured)

```
[3 4]
```

Live capture from go run ./cmd/runall on local ../welvet (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/04-weights).

Live output

```
[3 4]
```

5. quant — 20 pack formats

suite=velvet slug=05-quant status=PASS time=0.04s

README

5. quant - 20 pack formats

Part: II - Foundation

Package: github.com/openfluke/velvet/quant

Status: ok - [OK]

When

You need the quant foundation package.

Where

```
import "github.com/openfluke/velvet/quant"
```

```
cd 05-quant && source ../env.sh && go run .
```

Why

Inference, storage, and train need classic Q-packs, k-quants, IQ, Ternary/Binary, and Affine without a separate QAT mode or retained f32 master. Format is storage truth.

What

Pack / Unpack / MatVec / MatVecT for FormatNone...AffinePacked. Dense SIMD once-projects codes into Int8QS + scales (EnsureQ* / EnsureK/IQ/AffineSIMDCache) - no full-matrix F32 inflate for k/IQ/Affine. SGD on packed stores: short-lived unpack scratch -> update -> re-Pack; Packed stays truth.

Sample output (captured)

```
rows 2 cols 4 y [0.6857143 -1.4901161e-08]
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/05-quant).

Live output

```
rows 2 cols 4 y [0.6857143 -1.4901161e-08]
```

6. simd — Plan 9 kernels

suite=velvet slug=06-simd status=PASS time=0.04s

README

6. simd - Plan 9 kernels

Part: II - Foundation

Package: github.com/openfluke/velvet/simd

Status: ok - [OK]

When

You need the simd foundation package.

Where

```
import "github.com/openfluke/velvet/simd"
```

```
cd 06-simd && source ../env.sh && go run .
```

Why

CPU peak needs hand-written AVX2/NEON without a silent Go fallback that pretends SIMD ran.

What

amd64/arm64 .s kernels: DotTile, DotI8/U8, DotQ4_0, Saxpy, BitNet helpers, packed f16/bf16/fp8/fp4 dots; amd64 AVX2 DotTileF64 (WireF64); Go fused DotKRow / DotIQRow / DotAffineRow for k/IQ/Affine. SimdEnabled() false -> BackendSIMD hard-errors.

Sample output (captured)

```
10
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/06-simd).

Live output

```
10
```

7. webgpu — device GEMV & shaders

suite=velvet slug=07-webgpu status=PASS time=0.28s

README

7. webgpu - device GEMV & shaders

Part: II - Foundation

Package: github.com/openfluke/welvet/webgpu

Status: ok - [OK]

When

You need the webgpu foundation package.

Where

import "github.com/openfluke/welvet/webgpu"

```
cd 07-webgpu && source ../env.sh && go run .
```

Why

GPU paths must bind a real adapter. Host "fake GPU" was banned so suites cannot stamp WebGPU done when ALU ran on CPU.

What

DenseGEMV family (incl. quant/I8/resident), DenseGEMVT/DenseDW, RMSNorm, LayerNorm fwd, Softmax family, SwiGLUFuse. Available()/InitError() gate use.

Sample output (captured)

```
webgpu probe goos=linux goarch=amd64 WELVET_WGPU_BACKEND=""
  adapter high-perf: name="NVIDIA GeForce GTX 1650 SUPER" vendor="NVIDIA" arch="" driver="610.57.04" backend=vulkan type=discrete-gpu | maxBuf=1099511627776 maxSSBO=2147483644 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
    -> selected high-perf
  adapter low-power: name="Intel(R) UHD Graphics 630 (CML GT2)" vendor="Intel open-source Mesa driver" arch="" driver="Mesa 26.1.8" backend=vulkan type=integrated-gpu | maxBuf=2147483647 maxSSBO=2147483644 maxSSBO/stage=103 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
  adapter force-fallback: name="llvmpipe (LLVM 22.1.8, 256 bits)" vendor="llvmpipe" arch="" driver="Mesa 26.1.8 (LLVM 22.1.8)" backend=vulkan type=cpu | maxBuf=2147483647 maxSSBO=134217728 maxSSBO/stage=48 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
  adapter default: name="NVIDIA GeForce GTX 1650 SUPER" vendor="NVIDIA" arch="" driver="610.57.04" backend=vulkan type=discrete-gpu | maxBuf=1099511627776 maxSSBO=2147483644 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
    try RequestDevice inflated: maxBuf=2147483648 maxSSBO=1073741824 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
RESULT: OK via=inflated adapter="NVIDIA GeForce GTX 1650 SUPER" backend=vulkan maxSSBO=2147483644 maxBuf=1099511627776
[1.5 2.5] <nil> NVIDIA GeForce GTX 1650 SUPER
```

Live capture from go run ./cmd/runall on local ../.. /welvet (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/07-webgpu).

Live output

```
webgpu probe goos=linux goarch=amd64 WELVET_WGPU_BACKEND=""
  adapter high-perf: name="NVIDIA GeForce GTX 1650 SUPER" vendor="NVIDIA" arch="" driver="610.57.04" backend=vulkan type=discrete-gpu | maxBuf=1099511627776 maxSSBO=2147483644 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
    -> selected high-perf
  adapter low-power: name="Intel(R) UHD Graphics 630 (CML GT2)" vendor="Intel open-source Mesa driver" arch="" driver="Mesa 26.1.8" backend=vulkan type=integrated-gpu | maxBuf=2147483647 maxSSBO=2147483644 maxSSBO/stage=103 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
  adapter force-fallback: name="llvmpipe (LLVM 22.1.8, 256 bits)" vendor="llvmpipe" arch="" driver="Mesa 26.1.8 (LLVM 22.1.8)" backend=vulkan type=cpu | maxBuf=2147483647 maxSSBO=134217728 maxSSBO/stage=48 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
  adapter default: name="NVIDIA GeForce GTX 1650 SUPER" vendor="NVIDIA" arch="" driver="610.57.04" backend=vulkan type=discrete-gpu | maxBuf=1099511627776 maxSSBO=2147483644 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
    try RequestDevice inflated: maxBuf=2147483648 maxSSBO=1073741824 maxSSBO/stage=524288 maxBindGroups=8 maxComputeWG=65535 maxInvoc/WG=1024
RESULT: OK via=inflated adapter="NVIDIA GeForce GTX 1650 SUPER" backend=vulkan maxSSBO=2147483644 maxBuf=1099511627776
[1.5 2.5] <nil> NVIDIA GeForce GTX 1650 SUPER
```

8. tiling — SC/MC & workgroups

suite=velvet slug=08-tiling status=PASS time=0.04s

README

8. tiling - SC/MC & workgroups

Part: II - Foundation

Package: github.com/openfluke/velvet/tiling

Status: ok - [OK]

When

You need the tiling foundation package.

Where

```
import "github.com/openfluke/velvet/tiling"
```

```
cd 08-tiling && source ../env.sh && go run .
```

Why

MatVec throughput depends on tile size and when to go multi-core vs GPU workgroups. Centralizing caps keeps Dense and friends consistent.

What

DefaultCPUTile, DefaultGPUWG, CPUTile, PreferMultiCore, GPUWorkgroupsX.

Sample output (captured)

```
32 true 16
```

Live capture from `go run ./cmd/runall` on local `../.. /velvet` (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/08-tiling).

Live output

```
32 true 16
```

9. architecture — volumetric grid

suite=welvet slug=09-architecture status=PASS time=0.04s

README

9. architecture - volumetric grid

Part: II - Foundation

Package: github.com/openfluke/welvet/architecture

Status: ok - [OK]

When

You need the architecture foundation package.

Where

```
import "github.com/openfluke/welvet/architecture"
```

```
cd 09-architecture && source ../env.sh && go run .
```

Why

Networks are spatial (Depth×Rows×Cols×LayersPerCell), not only linear stacks. Topology lives here; compute lives in layer packages.

What

Grid/Cell/Coord, NewGrid, BindOp, HopOrder, SetRemoteLink/ResolveHop. VolumetricNetwork is an alias for Grid.

Sample output (captured)

```
cells 2
```

Live capture from `go run ./cmd/runall` on local `../..../welvet` (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/09-architecture).

Live output

```
cells 2
```

10. fusedgpu — decoder on device

suite=velvet slug=10-fusedgpu status=PASS time=0.06s

README

10. fusedgpu - decoder on device

Part: II - Foundation

Package: github.com/openfluke/velvet/fusedgpu

Status: ok - [OK]

When

You need the fusedgpu foundation package.

Where

```
import "github.com/openfluke/velvet/fusedgpu"
```

```
cd 10-fusedgpu && source ../env.sh && go run .
```

Why

Token-by-token host round-trips kill decode. A fused engine keeps weights and scratch resident for Q4_0 and BinaryG128 hybrid paths.

What

Engine from Spec (AppendTokens/Reset/Close); HybridEngine (PrefillSample/DecodeSample/DecodeChunk). Specs come from model/transformer export.

Sample output (captured)

```
need an ENTITY file: open model.entity: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/10-fusedgpu).

Live output

```
need an ENTITY file: open model.entity: no such file or directory
```

11. layers/dense — MatVec microkernel

suite=velvet slug=11-dense status=PASS time=0.04s

README

11. layers/dense - MatVec microkernel

Part: III - Layers

Package: github.com/openfluke/velvet/layers/dense

Status: ok - [OK]

When

Building or training a net that needs the layers/dense Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/dense"
```

```
cd 11-dense && source ../env.sh && go run .
```

Why

Most FLOPs are $W @ x$. One Dense stack owns $\text{FormatNone} \times 34$ and all quants $\times$ three backends so every composite proj shares one correctness surface - including native in-dtype SGD.

What

New / NewConfigured[T], Forward/Backward (dispatch on Exec.Backend), Place, ApplyGradSGD ($\rightarrow$ weights.ApplySGD on the store). SIMD forward (v1.0.3): dtype switch by MatVec strategy - DotTile / DotI8 / lowp packed / expand-once $\rightarrow$ DotTile / WireF64+DotTileF64; fused Dot* for classic Q*, k/IQ, AffinePacked. Composites (MHA, SwiGLU, CNN im2col, RNN/LSTM/Mamba, residual-sequential-parallel) reuse Dense children via syncProjExec. BackwardSIMD still DecodeRow/saxpy (not the new expand wires).

Sample output (captured)

```
8 32
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/11-dense).

Live output

```
8 32
```

12. layers/mha — attention

suite=velvet slug=12-mha status=PASS time=0.04s

README

12. layers/mha - attention

Part: III - Layers

Package: github.com/openfluke/velvet/layers/mha

Status: ok - [OK]

When

Building or training a net that needs the layers/mha Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/mha"
```

```
cd 12-mha && source ../env.sh && go run .
```

Why

Transformers need multi-head attention with masks, RoPE/ALiBi, GQA/MQA, and cross-attn - without forking MatVec for every projection.

What

Q/K/V/O are Dense children. Presets: DecoderCausal, EncoderBidirectional, CrossAttention, ... Attn/RoPE ALU is host today; on-device attn shaders are still open.

Sample output (captured)

```
true 256 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/12-mha).

Live output

```
true 256 <nil>
```

13. layers/swiglu — gated FFN

suite=velvet slug=13-swiglu status=PASS time=0.05s

README

13. layers/swiglu - gated FFN

Part: III - Layers

Package: github.com/openfluke/velvet/layers/swiglu

Status: ok - [OK]

When

Building or training a net that needs the layers/swiglu Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/swiglu"
```

```
cd 13-swiglu && source ../env.sh && go run .
```

Why

Modern decoder FFNs are SiLU(gate)up -> down. Projections must share Dense's quant/backend matrix.

What

Gate/Up/Down Dense children; DefaultFFN(dModel); WebGPU SiLU fuse on forward.

Sample output (captured)

```
64 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/13-swiglu).

Live output

```
64 <nil>
```

14. layers/rmsnorm

suite=velvet slug=14-rmsnorm status=PASS time=0.06s

README

14. layers/rmsnorm

Part: III - Layers

Package: github.com/openfluke/velvet/layers/rmsnorm

Status: ok - [OK]

When

Building or training a net that needs the layers/rmsnorm Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/rmsnorm"
```

```
cd 14-rmsnorm && source ../env.sh && go run .
```

Why

Llama-style blocks normalize by RMS, not mean+var. Needs native fwd/bwd and WebGPU shaders.

What

Per-token RMS + γ on weights.Store; WebGPU fwd+bwd; SIMD DotTile stats + host scale.

Sample output (captured)

```
[0 0 0 0] <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/14-rmsnorm).

Live output

```
[0 0 0 0] <nil>
```

15. layers/layernorm

suite=velvet slug=15-layernorm status=PASS time=0.05s

README

15. layers/layernorm

Part: III - Layers

Package: github.com/openfluke/velvet/layers/layernorm

Status: ok - [OK]

When

Building or training a net that needs the layers/layernorm Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/layernorm"
```

```
cd 15-layernorm && source ../env.sh && go run .
```

Why

Classic mean+var normalization with γ/β - still required for many HF architectures.

What

WebGPU forward; backward host today. Same dtype×quant axes as other weighted layers.

Sample output (captured)

```
8 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/15-layernorm).

Live output

```
8 <nil>
```

16. layers/embedding

suite=velvet slug=16-embedding status=PASS time=0.05s

README

16. layers/embedding

Part: III - Layers

Package: github.com/openfluke/velvet/layers/embedding

Status: ok - [OK]

When

Building or training a net that needs the layers/embedding Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/embedding"
```

```
cd 16-embedding && source ../env.sh && go run .
```

Why

Token IDs must gather rows from a table - not a Dense MatVec - with scatter grads on backward.

What

Config{VocabSize, EmbeddingDim, SeqLen}; table on weights.Store; host gather on SIMD/WebGPU today.

Sample output (captured)

```
[1 4 16] <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/16-embedding).

Live output

```
[1 4 16] <nil>
```

17. layers/softmax

suite=welvet slug=17-softmax status=PASS time=0.05s

README

17. layers/softmax

Part: III - Layers

Package: github.com/openfluke/welvet/layers/softmax

Status: ok - [OK]

When

Building or training a net that needs the layers/softmax Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/welvet/layers/softmax"
```

```
cd 17-softmax && source ../env.sh && go run .
```

Why

Classification heads and attention need stable softmax variants, including sparse/Gumbel/Entmax for research paths.

What

Weightless layer; KindStandard/Temperature/Grid/Hierarchical/Gumbel/Masked/Sparse/... WebGPU covers std family; exotic kinds hard-error on GPU (no silent host).

Sample output (captured)

```
[0.6380664 0.23473153 0.09543471 0.031767454] <nil>
```

Live capture from `go run ./cmd/runall` on local `../..../welvet` (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/17-softmax).

Live output

```
[0.6380664 0.23473153 0.09543471 0.031767454] <nil>
```

18. layers/sequential

suite=velvet slug=18-sequential status=PASS time=0.05s

README

18. layers/sequential

Part: III - Layers

Package: github.com/openfluke/velvet/layers/sequential

Status: ok - [OK]

When

Building or training a net that needs the layers/sequential Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/sequential"
```

```
cd 18-sequential && source ../env.sh && go run .
```

Why

Some cells need an ordered Dense chain without burning grid hops.

What

Dense->Dense compose, or mixed Ops via NewFromOps (Dense, SwiGLU, RMSNorm, LayerNorm). Parallel as a child is parallel.ResidualGraft / a Sandwich - Sequential cannot import parallel.

Sample output (captured)

```
16 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/18-sequential).

Live output

```
16 <nil>
```

19. layers/residual

suite=velvet slug=19-residual status=PASS time=0.05s

README

19. layers/residual

Part: III - Layers

Package: github.com/openfluke/velvet/layers/residual

Status: ok - [OK]

When

Building or training a net that needs the layers/residual Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/residual"
```

```
cd 19-residual && source ../env.sh && go run .
```

Why

Skip connections stabilize deep stacks: $y = F(x) + x$ with correct skip grads.

What

F is Dense Dim->Dim, or mixed Ops via NewFromOps (Dense, SwiGLU, RMSNorm, LayerNorm). Parallel as F is parallel.ResidualGraft ($y = F(x)+x$) - Residual cannot import parallel.

Sample output (captured)

```
16 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/19-residual).

Live output

```
16 <nil>
```

20. layers/cnn1 · cnn2 · cnn3

suite=velvet slug=20-cnn status=PASS time=0.05s

README

20. CNN family - cnn1 / cnn2 / cnn3

When: spatial / temporal convolution senses (1D seq, 2D images, 3D volumes).

Where: layers/cnn1, layers/cnn2, layers/cnn3

Why: im2col -> Dense Proj so quant/SIMD/WebGPU and CamSync (via Proj) are shared.

```
cd 20-cnn && source ../env.sh && go run .
```

Op | Use

cnn1 | Audio / 1D sequences

cnn2 | Images / MNIST-style

cnn3 | Volumetric / video voxels

Put CNN inside Parallel cams when you want CamSync on kernels.

Live output

```
cnn1 [1 4 14]
```

```
cnn2 [1 4 6 6]
```

```
cnn3 [1 2 3 3 3]
```

21. layers/rnn · lstm

suite=velvet slug=21-rnn-lstm status=PASS time=0.05s

README

21. layers/rnn - lstm

Part: III - Layers

Package: github.com/openfluke/velvet/layers/lstm

Status: ok - [OK]

When

Building or training a net that needs the layers/lstm Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/lstm"
```

```
cd 21-rnn-lstm && source ../env.sh && go run .
```

Why

Sequence models before transformers still need vanilla RNN and LSTM with BPTT on the shared MatVec stack.

What

IH/HH (and LSTM gates) via Dense; recurrence ALU host; device required for WebGPU path.

Sample output (captured)

```
64 64
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/21-rnn-lstm).

Live output

```
64 64
```

22. layers/seqmix — mixer contract

suite=velvet slug=22-seqmix status=PASS time=0.04s

README

22. layers/seqmix - mixer contract

Part: III - Layers

Package: github.com/openfluke/velvet/layers/seqmix

Status: ok - [OK]

When

Building or training a net that needs the layers/seqmix Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/seqmix"
```

```
cd 22-seqmix && source ../env.sh && go run .
```

Why

Attention, SSM, linear attn, and conv mixers must not be accidental forks of mha. Naming the contract keeps packages honest.

What

KindAttention | KindSSM | KindLinearAttn | KindConvMix and Contract{Kind,DModel,MaxT}. No compute here.

Sample output (captured)

```
attention
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/22-seqmix).

Live output

```
attention
```

23. layers/gdn — gated delta net

suite=velvet slug=23-gdn status=PASS time=0.05s

README

23. layers/gdn - gated delta net

Part: III - Layers

Package: github.com/openfluke/velvet/layers/gdn

Status: ok - [OK]

When

Building or training a net that needs the layers/gdn Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/gdn"
```

```
cd 23-gdn && source ../env.sh && go run .
```

Why

Linear attention / decode-first mixers (Gated DeltaNet) need a first-class package under KindLinearAttn.

What

Exec CPU/SIMD/WebGPU; ForwardDecode; truncated BPTT. Timed matrix is Float32-primary: PermutationOK is f32 × FormatNone/BinaryPacked × CPU/SIMD/WebGPU; other cells GAP (declared), not FAIL.

Sample output (captured)

```
<nil>
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/23-gdn).

Live output

```
<nil>
```

24. layers/mamba — selective SSM

suite=velvet slug=24-mamba status=PASS time=0.05s

README

24. layers/mamba - selective SSM

Part: III - Layers

Package: github.com/openfluke/velvet/layers/mamba

Status: ok - [OK]

When

Building or training a net that needs the layers/mamba Op (also usable as a Parallel cam).

Where

import "github.com/openfluke/velvet/layers/mamba"

```
cd 24-mamba && source ../env.sh && go run .
```

Why

SSM mixers (KindSSM) are not MHA clones - they need their own selective-scan path.

What

InProj -> softplus(Δ) scan -> OutProj. Full timed matrix + train grids (scan ALU host).

Sample output (captured)

```
true <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/24-mamba).

Live output

```
true <nil>
```

25. layers/convt1 · convt2 · convt3

suite=velvet slug=25-convt status=PASS time=0.07s

README

25. ConvTranspose family - convt1 / convt2 / convt3

When: upsampling / decoder paths (1D, 2D, 3D).

Where: layers/convt1, layers/convt2, layers/convt3

Why: generative twins and U-Net-style expands; same Proj/Dense backend story as CNN.

```
cd 25-convt && source ../env.sh && go run .
```

Live output

```
convt1 [1 2 17]
convt2 [1 2 5 5]
convt3 [1 2 3 3 3]
```

26. layers/kmeans

suite=velvet slug=26-kmeans status=PASS time=0.05s

README

26. layers/kmeans

Part: III - Layers

Package: github.com/openfluke/velvet/layers/kmeans

Status: ok - [OK]

When

Building or training a net that needs the layers/kmeans Op (also usable as a Parallel cam).

Where

import "github.com/openfluke/velvet/layers/kmeans"

```
cd 26-kmeans && source ../env.sh && go run .
```

Why

Soft clustering as a differentiable layer lets topology experiments sit inside the same train loop.

What

Centers on Dense ($K \times \text{FeatureDim}$); soft assignment outputs. Full timed matrix + train grids.

Sample output (captured)

```
[1 4] <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/26-kmeans).

Live output

```
[1 4] <nil>
```

27. layers/parallel — MoE + cameral

suite=velvet slug=27-parallel status=PASS time=0.05s

README

27. layers/parallel - MoE + cameral

Part: III - Layers

Package: github.com/openfluke/velvet/layers/parallel

Status: ok - [OK]

When

Building or training a net that needs the layers/parallel Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/parallel"
```

```
cd 27-parallel && source ../env.sh && go run .
```

Why

Mixture-of-experts and multi-path cells need concat/add/avg/filter combines. Cameral graphs need sibling hemispheres that share input, merge outputs, and optionally train under distinct modes.

What

Parallel branches + Stack sandwiches. Hemispheres / Bicameral / Sandwich build nested multi-cameral nets. SetBranchModes + TrainStackMSE let each hemi use a different TrainMode (all 29 named updates - BP, Tween, Split, FastProxy, Sparse, Step*, Mesh*, ...).

Sample output (captured)

```
loss 0 err <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/27-parallel).

Live output

```
loss 0 err <nil>
```

28. layers/metacognition

suite=velvet slug=28-metacognition status=PASS time=0.05s

README

28. layers/metacognition

Part: III - Layers

Package: github.com/openfluke/velvet/layers/metacognition

Status: ok - [OK]

When

Building or training a net that needs the layers/metacognition Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/metacognition"
```

```
cd 28-metacognition && source ../env.sh && go run .
```

Why

Observed layers can apply heuristic stability rules (gate/scale/reset) without dtype morph/QAT.

What

Wraps Dense + DefaultStabilityRules(); Stats exposed. Full timed matrix + train grids.

Sample output (captured)

```
16 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/28-metacognition).

Live output

```
16 <nil>
```

29. runtime/forward

suite=velvet slug=29-forward status=PASS time=0.07s

README

29. runtime/forward

Part: IV - Runtime

Package: github.com/openfluke/velvet/runtime/forward

Status: ok - [OK]

When

Walking a Grid/Stack with the shared runtime/forward path.

Where

```
import "github.com/openfluke/velvet/runtime/forward"
```

```
cd 29-forward && source ../env.sh && go run .
```

Why

A grid of heterogeneous ops needs one walker that dispatches by concrete type and fails loudly on unknowns.

What

ForwardT -> Result tape; Cell[T] for single-cell dispatch. Covers Dense...Residual + extended wired ops.

Sample output (captured)

```
true <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/29-forward).

Live output

```
true <nil>
```

30. runtime/backward

suite=velvet slug=30-backward status=PASS time=0.26s

README

30. runtime/backward

Part: IV - Runtime

Package: github.com/openfluke/velvet/runtime/backward

Status: ok - [OK]

When

Walking a Grid/Stack with the shared runtime/backward path.

Where

```
import "github.com/openfluke/velvet/runtime/backward"
```

```
cd 30-backward && source ../env.sh && go run .
```

Why

Training needs a reverse tape over the same ops forward used - no separate graph framework.

What

BackwardT walks the tape and returns per-op weight grads.

Sample output (captured)

```
true <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/30-backward).

Live output

```
true <nil>
```

31. runtime/training

suite=velvet slug=31-training status=PASS time=0.08s

README

31. runtime/training

Part: IV - Runtime

Package: github.com/openfluke/velvet/runtime/training

Status: ok - [OK]

When

Walking a Grid/Stack with the shared runtime/training path.

Where

```
import "github.com/openfluke/velvet/runtime/training"
```

```
cd 31-training && source ../env.sh && go run .
```

Why

Suites and small nets need MSE+SGD and tween hooks without inventing an external trainer or a retained float32 master beside storage.

What

MSE/MSEGrad, SGD, Step, ApplyTween/StepTween, StepMesh. Layer-agnostic ApplyGradSGD dispatch (Dense...Mamba/GDN/...). FormatNone: in-dtype ApplySGD; packed: unpack->update->re-Pack. No QAT dual path - storage dtype/format is truth after every step. Sandwich credit (Split / FastProxy / Sparse / ...) lives in layers/parallel TrainMode - see §67.

Sample output (captured)

```
0 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/31-training).

Live output

```
0 <nil>
```

32. runtime/step — step mesh

suite=velvet slug=32-step status=PASS time=0.06s

README

32. runtime/step - step mesh

Part: IV - Runtime

Package: github.com/openfluke/velvet/runtime/step

Status: ok - [OK]

When

Walking a Grid/Stack with the shared runtime/step path.

Where

```
import "github.com/openfluke/velvet/runtime/step"
```

```
cd 32-step && source ../env.sh && go run .
```

Why

Spatial feedback (remote links) needs a discrete-time mesh where every cell updates from a double buffer - different from a decoder wavefront. Cross-numeric train also needs the same mesh with weight DType activation Tensor[T].

What

State[T], StepForward/StepBackward/StepApplyTween / StepMesh across the grid for all wired Ops × dtype × quant × CPU/SIMD. W2A Cross-Numeric Train: polyops.AllKinds() × weight dtype × act host (smoke $\sim 21 \times 7 \times 5 \approx 735$; full $\sim 21 \times 34 \times 15 \approx 10.7k$) - asserts no retained f32 master after StepMesh. Stack Step* is a different clock: IsLineStep / TrainLine (1D pipe) - see §67.

Sample output (captured)

```
<nil>
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/32-step).

Live output

```
<nil>
```

65. Cross-numeric train + down-the-dem

suite=velvet slug=65-cross-numeric status=PASS time=0.05s

README

65. Cross-numeric train + down-the-dem

Part: IV - Runtime

Package: github.com/openfluke/velvet/runtime/training

Status: ok - [OK]

When

Walking a Grid/Stack with the shared runtime/training path.

Where

```
import "github.com/openfluke/velvet/runtime/training"
```

```
cd 65-cross-numeric && source ../env.sh && go run .
```

Why

Weight storage dtype and activation Tensor[T] are independent axes. Proving train without a retained float32 master means sweeping $W \times A$ - not only matched float32 acts.

What

W2A Step Cross-Numeric Train: polyops.AllKinds() $\times$ FormatNone weight dtype $\times$ Go Numeric act host (smoke ~735; full ~10.7k) via StepMesh, then assert no retained f32 master. Public Dense volumetric showcase: down-the-dem - dtype demotion ladder, packed quants, and the full $34 \times 15 \times 3$ perm matrix with charts/PDF.

Sample output (captured)

```
1.5 <nil> false
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/65-cross-numeric).

Live output

```
1.5 <nil> false
```

67. TrainMode — 29 named updates

suite=velvet slug=67-train-modes status=PASS time=0.05s

README

67. TrainMode - 29 named updates

Part: IV - Runtime

Package: github.com/openfluke/velvet/layers/parallel

Status: ok - [OK] 29 modes

When

Building or training a net that needs the layers/parallel Op (also usable as a Parallel cam).

Where

import "github.com/openfluke/velvet/layers/parallel"

```
cd 67-train-modes && source ../env.sh && go run .
```

Why

Backprop is one update, not the only one. Credit assignment (broadcast gap, head proxy, sparse duty clock) has to be a named axis you can race - not a comment in a notebook. Cameral Mix also needs one TrainMode per hemisphere on the same loss.

What

parallel.TrainMode: AllNamedTrainModes() = 29 (Inherit omitted). Stack-local Split/Alt plus Step* 1D pipe twins and Mesh* grid schedulers. TrainStackMSE / TrainStackCE honour BranchModes. Display names use Short() / ShortTrainMode. Rival metric is hard Acc vs StepBP; Lucy Score is Tput × Avail × Acc - do not mix those sentences.

Sample output (captured)

```
named 31 linestep 11
stepfastproxy Step[T][S][FP] <nil>
[T]=Tween [S]=Split [FP]=FastProxy [L]=Linear [HP]=HeadProxy [F]=Freeze [Sh]=Shadow [A]=Adv [M]=Memory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/67-train-modes).

Live output

```
named 31 linestep 11
stepfastproxy Step[T][S][FP] <nil>
[T]=Tween [S]=Split [FP]=FastProxy [L]=Linear [HP]=HeadProxy [F]=Freeze [Sh]=Shadow [A]=Adv [M]=Memory
```

33. systems/dna

suite=velvet slug=33-dna status=PASS time=0.05s

README

33. systems/dna

Part: V - Systems

Package: github.com/openfluke/velvet/systems/dna

Status: ok - [OK]

When

Adaptation, measurement, or evolution (systems/dna).

Where

import "github.com/openfluke/velvet/systems/dna"

```
cd 33-dna && source ../env.sh && go run .
```

Why

Quant and train must be measurable as topology/weight fingerprints - DNA detects logic shifts.

What

ExtractDNA, CompareNetworks, CosineSimilarity, FlattenOp / CollectStores across dtype×quant.

Sample output (captured)

```
{1 map[0,0,0,0:1] []}
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/33-dna).

Live output

```
{1 map[0,0,0,0:1] []}
```

34. systems/evolution

suite=welvet slug=34-evolution status=PASS time=0.05s

README

34. systems/evolution

Part: V - Systems

Package: github.com/openfluke/welvet/systems/evolution

Status: ok - [OK]

When

Adaptation, measurement, or evolution (systems/evolution).

Where

```
import "github.com/openfluke/welvet/systems/evolution"
```

```
cd 34-evolution && source ../env.sh && go run .
```

Why

Topology search and weight crossover need first-class splice + NEAT on CPU-resident grids.

What

SpliceDNA, NEATMutate, NewNEATPopulation, CloneGrid - dtype/quant preserved via SetFromF32.

Sample output (captured)

```
<nil>
```

Live capture from go run ./cmd/runall on local ../../welvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/34-evolution).

Live output

```
<nil>
```

35. systems/tween

suite=velvet slug=35-tween status=PASS time=0.06s

README

35. systems/tween

Part: V - Systems

Package: github.com/openfluke/velvet/systems/tween

Status: ok - [OK]

When

Adaptation, measurement, or evolution (systems/tween).

Where

```
import "github.com/openfluke/velvet/systems/tween"
```

```
cd 35-tween && source ../env.sh && go run .
```

Why

Target propagation (chain-rule or Hebbian layerwise gaps) is an alternative credit-assignment path. Not the same package as TrainMode Tween / TweenChain on a Sandwich (those are layers/parallel - §67).

What

NewState, Forward, BackwardChainRule / BackwardLayerwise, ApplyGaps; SIMD DotTile/Saxpy budgets.

Sample output (captured)

```
forward: no op at {0 0 0 0} (type Dense)
```

Live capture from go run ./cmd/runall on local ../..../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/35-tween).

Live output

```
forward: no op at {0 0 0 0} (type Dense)
```

36. systems/tanhi — UDP HUD

suite=velvet slug=36-tanhi status=PASS time=0.05s

README

36. systems/tanhi - UDP HUD

Part: V - Systems

Package: github.com/openfluke/velvet/systems/tanhi

Status: ok - [OK]

When

Adaptation, measurement, or evolution (systems/tanhi).

Where

```
import "github.com/openfluke/velvet/systems/tanhi"
```

```
cd 36-tanhi && source ../env.sh && go run .
```

Why

Training visualization must never block the engine - best-effort UDP JSON-lines to a HUD.

What

ConfigFromGrid, Emit/EmitSweep, DefaultUDPPort. SoulGlitch-style consumers.

_No captured output in _manifest.json yet - run go run . and paste here._

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/36-tanhi).

Sample output (captured)

```
- 36-tanhi (996ms)
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Live output

(no output)

37. systems/telemetry

suite=velvet slug=37-telemetry status=PASS time=0.05s

README

37. systems/telemetry

Part: V - Systems

Package: github.com/openfluke/velvet/systems/telemetry

Status: ok - [OK]

When

Adaptation, measurement, or evolution (systems/telemetry).

Where

import "github.com/openfluke/velvet/systems/telemetry"

```
cd 37-telemetry && source ../env.sh && go run .
```

Why

Static structural blueprints (sizes, op kinds) differ from live tanhi events.

What

ExtractNetworkBlueprint, ExtractLayerTelemetry for introspection/UIs.

Sample output (captured)

```
{ID:demo TotalLayers:1 TotalParams:0 Layers:[{Z:0 Y:0 X:0 L:0 Type:Dense Activation:ReLU Parameters:0 InputShape:[] OutputShape:[] Branches:[] CombineMode:}]}
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/37-telemetry).

Live output

```
{ID:demo TotalLayers:1 TotalParams:0 Layers:[{Z:0 Y:0 X:0 L:0 Type:Dense Activation:ReLU Parameters:0 InputShape:[] OutputShape:[] Branches:[] CombineMode:}]}
```

66. lucy — SoftAcc / Score measuring

suite=velvet slug=66-lucy status=PASS time=0.04s

README

66. lucy - SoftAcc / Score measuring

Part: V - Systems

Package: github.com/openfluke/velvet/lucy

Status: ok - [OK]

When

Adaptation, measurement, or evolution (lucy).

Where

```
import "github.com/openfluke/velvet/lucy"
```

```
cd 66-lucy && source ../env.sh && go run .
```

Why

Adaptation benches (test41-w, tide, live_gpt) need one shared measuring math - SoftAcc, Availability, AdaptPct, Score - not three copies of the formulas.

What

Pure measuring package: SoftAcc / SoftAccProb, Window + Snapshot, Finalize. No datasets, no train loops. Sine scale 0.10; classification SoftAccProb scale 1.0. Density / synthetic-organism board is §69.

Sample output (captured)

```
soft=20.0 class=91.0 avail=80.0 score=7680
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/66-lucy).

Live output

```
soft=20.0 class=91.0 avail=80.0 score=7680
```

69. Lucy density — synthetic organism

suite=welvet slug=69-lucy-density status=PASS time=0.04s

README

69. Lucy density - synthetic organism

Part: V - Systems

Package: github.com/openfluke/welvet/lucy

Status: ok - [OK] BuildLPD

When

Adaptation, measurement, or evolution (lucy).

Where

import "github.com/openfluke/welvet/lucy"

```
cd 69-lucy-density && source ../env.sh && go run .
```

Why

A new host (char LM, MNIST, a layer sprint) should not copy tide's goldilocks math. The question is always the same: can the net run and train at the same time in a small box, then how far that live-fit condenses without becoming a trap. That ruler belongs in the engine.

What

lucy.BuildLPD ranks Samples for consciousness (Acc / Throughput / Availability keep vs learner peaks) then Lucy density ($Q \times \text{shrink}$ vs Acc-champ RAM). Gold / near / keep / trap bands, consciousness radar, and memory-density radar are computed here. Tide dash and Lucy PDF only draw the board.

Sample output (captured)

```
lead=int8 band=gold LPD=5.11 trap=bin
```

Live capture from go run ./cmd/runall on local ../../welvet (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/69-lucy-density).

Live output

```
lead=int8 band=gold LPD=5.11 trap=bin
```

38. model/entity — .entity files

suite=velvet slug=38-entity status=PASS time=0.05s

README

38. model/entity - .entity files

Part: VI - Model IO

Package: github.com/openfluke/velvet/model/entity

Status: ok - [OK]

When

Checkpoints, HF import, tokenize, sample, or decode (model/entity).

Where

```
import "github.com/openfluke/velvet/model/entity"
```

```
cd 38-entity && source ../env.sh && go run .
```

Why

HF safetensors are awkward for native topology + packed weights. ENTITY is the Velvet checkpoint.

What

Open/Inspect/IsEntity, LoadBlob/LoadQuantBlob, PackFromHF/ImportFromHF, WriteTransformerFile, SerializeNetwork, WriteCameralFile / LoadCameral (sandwich Stack + TrainMode).

Sample output (captured)

```
open model.entity: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/38-entity).

Live output

```
open model.entity: no such file or directory
```

39. model/hf — snapshots

suite=velvet slug=39-hf status=PASS time=0.04s

README

39. model/hf - snapshots

Part: VI - Model IO

Package: github.com/openfluke/velvet/model/hf

Status: ok - [OK]

When

Checkpoints, HF import, tokenize, sample, or decode (model/hf).

Where

import "github.com/openfluke/velvet/model/hf"

```
cd 39-hf && source ../env.sh && go run .
```

Why

Import starts with probing HF/MLX layouts before packing ENTITY.

What

InspectSnapshot, DetectArchitecture, safetensors/MLX loaders, Qwen3.5 hybrid helpers.

Sample output (captured)

```
hf: config.json: stat /path/to/hf-snapshot/config.json: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/39-hf).

Live output

```
hf: config.json: stat /path/to/hf-snapshot/config.json: no such file or directory
```

40. model/tokenizer

suite=velvet slug=40-tokenizer status=PASS time=0.05s

README

40. model/tokenizer

Part: VI - Model IO

Package: github.com/openfluke/velvet/model/tokenizer

Status: ok - [OK]

When

Checkpoints, HF import, tokenize, sample, or decode (model/tokenizer).

Where

import "github.com/openfluke/velvet/model/tokenizer"

```
cd 40-tokenizer && source ../env.sh && go run .
```

Why

Generate needs encode/decode of HF tokenizer.json without pulling Python.

What

LoadTokenizer, Encode/Decode, LoadForEntity.

Sample output (captured)

```
failed to read tokenizer file: open tokenizer.json: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/40-tokenizer).

Live output

```
failed to read tokenizer file: open tokenizer.json: no such file or directory
```

41. model/sampling

suite=velvet slug=41-sampling status=PASS time=0.04s

README

41. model/sampling

Part: VI - Model IO

Package: github.com/openfluke/velvet/model/sampling

Status: ok - [OK]

When

Checkpoints, HF import, tokenize, sample, or decode (model/sampling).

Where

```
import "github.com/openfluke/velvet/model/sampling"
```

```
cd 41-sampling && source ../env.sh && go run .
```

Why

Logits -> token ID needs ArgMax, TopK+temperature, penalties, and chat hygiene in one place.

What

ArgMax, SampleTopK, ApplyRepetitionPenalty, BanIDs, SanitizeChatReply.

Sample output (captured)

```
1
1
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/41-sampling).

Live output

```
1
1
```

42. model/transformer — generate

suite=velvet slug=42-transformer status=PASS time=0.06s

README

42. model/transformer - generate

Part: VI - Model IO

Package: github.com/openfluke/velvet/model/transformer

Status: ok - [OK]

When

Checkpoints, HF import, tokenize, sample, or decode (model/transformer).

Where

```
import "github.com/openfluke/velvet/model/transformer"
```

```
cd 42-transformer && source ../env.sh && go run .
```

Why

ENTITY packs must run as Llama-style decoders with KV cache, profiles (SIMD/WebGPU/fused), and chat templates.

What

LoadEntity -> Model; Generate; ApplyExec profiles; ExportFusedGPUSpec / hybrid sync.

Sample output (captured)

```
open model.entity: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/42-transformer).

Live output

```
open model.entity: no such file or directory
```

43. apps — octo · flux2 · mosstts

suite=velvet slug=43-apps status=PASS time=0.04s

README

43. apps - octo - flux2 - mosstts

Part: VII - Apps

Package: [github.com/openfluke/velvet/apps/...](https://github.com/openfluke/velvet/apps/)

Status: partial - ?

When

You need the apps/... foundation package.

Where

import "github.com/openfluke/velvet/apps/..."

```
cd 43-apps && source ../env.sh && go run .
```

Why

Products must not pollute engine packages. Octo is the model shell; flux2/mosstts are domain apps. Lucy races (AAI test41 / test48 / test50) are benches, not Velvet packages.

What

octo (own module): download/convert/chat, see §44. flux2: MMDiT image. mosstts: Speak/SpeakToFile pipeline. AAI sandwiches race TrainMode on toys - see §68. Off the v1.0 engine board.

Sample output (captured)

```
Apps import the engine - the engine never imports apps.
  cd apps/octo && go run .
  cd apps/flux2 && go run .
  cd apps/mosstts && go run .
```

Live capture from go run ./cmd/runall on local ../..velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/43-apps).

Live output

```
Apps import the engine - the engine never imports apps.
Full when/where/why inventory: go run ../75-apps-map
apps root: /home/openfluke/git/velvet/apps (10 entries)
- aai
- biotalk
- flux2
- kokoro
- mosstts
- octo
- planetbridging
- qwenasr
- qwentts
```

44. Octo — model shell

suite=velvet slug=44-octo status=PASS time=0.04s

README

44. Octo - model shell

Part: VII - Apps

Package: github.com/openfluke/velvet/apps/octo

Status: ok - [OK] runs

When

You need the apps/octo foundation package.

Where

```
import "github.com/openfluke/velvet/apps/octo"
```

```
cd 44-octo && source ../env.sh && go run .
```

Why

A model is only useful with a shell around it: pull weights from Hugging Face, convert them to a Velvet .entity, then chat, serve, or benchmark. Octo is that shell, kept in its own module so the engine never depends on an app.

What

Subcommands cover the whole loop: hub download/ensure repos, convert pack to a single .entity, interactive run/serve/chat, plus image and speech menus. Its bench harness sweeps every quant format across a CPU Plan 9 SIMD fused profile and a WebGPU fused profile.

Sample output (captured)

```
Octo is its own module - the engine never imports it.  
cd apps/octo && go run .  
download -> convert -> .entity -> chat / serve / bench
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/44-octo).

Live output

```
Octo is its own module - the engine never imports it.  
cd apps/octo && go run .  
download -> convert -> .entity -> chat / serve / bench
```

68. Cameral sandwiches + AAI Lucy

suite=velvet slug=68-cameral status=PASS time=0.05s

README

68. Cameral sandwiches + AAI Lucy

Part: VII - Apps

Package: github.com/openfluke/velvet/layers/parallel

Status: ok - [OK] cameral

When

Building or training a net that needs the layers/parallel Op (also usable as a Parallel cam).

Where

```
import "github.com/openfluke/velvet/layers/parallel"
```

```
cd 68-cameral && source ../env.sh && go run .
```

Why

A single Dense chain cannot host two independent weight copies that share an input, merge, and optionally train under different updates. That is the cameral graph: hemispheres, not screen-space sprites and not a second hidden size.

What

Hemispheres / Bicameral / Sandwich / Mix. Stem -> Parallel mid -> Dense head. SetBranchModes + TrainStackMSE. Lucy races in AAI (test41 / test48 / test50) import this API; measuring is lucy/; harness is not engine.

Sample output (captured)

```
mix loss 0 err <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/68-cameral).

Live output

```
mix loss 0 err <nil>
```

70. CamSync — inter-cameral / cross-mesh weight blend

suite=velvet slug=70-cam-sync status=PASS time=0.05s

README

70. CamSync - inter-cameral / cross-mesh weight blend

Part: VII - Apps

Package: github.com/openfluke/velvet/layers/parallel

Status: ok - [OK] CamSync

When

Building or training a net that needs the layers/parallel Op (also usable as a Parallel cam).

Where

import "github.com/openfluke/velvet/layers/parallel"

```
cd 70-cam-sync && source ../env.sh && go run .
```

Why

Mix BranchModes and different inits let cams diverge on purpose. Sometimes you want them to share - gently (1% pull) or hard (full average) - within a Parallel, across Stack children, or even between same-shaped stores sitting on wildly different mesh layouts ($1 \times 2 \times 1 \leftrightarrow 2 \times 4 \times 5$). That is CamSync: couple weights without collapsing the graph into one Dense.

What

CamSyncConfig{Alpha, When, Groups, Cross}. BlendStores pulls every store in a clique toward the mean. Empty Groups = all cams; Cross wires SyncEndpoint pairs across layers/cams. Shape rule: Rows×Cols must match. Bidirectional today; one-way teacher->student not yet.

Sample output (captured)

```
cosine before=-0.0697 after=1.0000
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/70-cam-sync).

Live output

```
cosine before=-0.0697 after=1.0000
```

45. stub/seed

suite=velvet slug=45-stub-seed status=PASS time=0.05s

README

45. stub/seed

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/seed

Status: partial - ?

When

Scaffold / experimental stub stub/seed (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/seed"
```

```
cd 45-stub-seed && source ../env.sh && go run .
```

Why

Ship topology recipes (layer seeds -> He-init) without weight blobs.

What

SeedFrom/From, RNG, InitStoreHe, Dense/MHA/SwiGLU/CNN infinite manifests.

Sample output (captured)

```
0.4118750632366808
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/45-stub-seed).

Live output

```
0.4118750632366808
```

46. stub/serialization

suite=velvet slug=46-stub-serialization status=PASS time=0.05s

README

46. stub/serialization

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/serialization

Status: partial - ?

When

Scaffold / experimental stub stub/serialization (may be partial).

Where

import "github.com/openfluke/velvet/stub/serialization"

```
cd 46-stub-serialization && source ../env.sh && go run .
```

Why

Volumetric grids need JSON/ENTITY persist beyond transformer packs.

What

SerializeEntity/LoadEntity, SerializeGrid/GridFromSpec - native FormatNone bytes; packed via wire.

Sample output (captured)

```
101 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/46-stub-serialization).

Live output

```
101 <nil>
```

47. stub/memory

suite=velvet slug=47-stub-memory status=PASS time=0.05s

README

47. stub/memory

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/memory

Status: partial - ?

When

Scaffold / experimental stub stub/memory (may be partial).

Where

import "github.com/openfluke/velvet/stub/memory"

```
cd 47-stub-memory && source ../env.sh && go run .
```

Why

HF->ENTITY and GPU upload need footprint accounting and optional history charts.

What

FromGrid, Footprint, InitScavenger, ReleaseTransient; WELVET_MEMORY_HISTORY=1.

Sample output (captured)

```
{HostWeightsMB:0 GPUWeightsMB:0 GPUKVMB:0}
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/47-stub-memory).

Live output

```
{HostWeightsMB:0 GPUWeightsMB:0 GPUKVMB:0}
```

48. stub/donate

suite=velvet slug=48-stub-donate status=PASS time=0.05s

README

48. stub/donate

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/donate

Status: partial - ?

When

Scaffold / experimental stub stub/donate (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/donate"
```

```
cd 48-stub-donate && source ../env.sh && go run .
```

Why

LAN donors should accept framed JSON jobs without embedding HTTP in the engine.

What

u32-LE + JSON frames; ServeTCP/Dial; model_push vs local_lm. v0 workers echo.

Sample output (captured)

```
default port 17001
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/48-stub-donate).

Live output

```
default port 17001
```

49. stub/fountain

suite=velvet slug=49-stub-fountain status=PASS time=0.06s

README

49. stub/fountain

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/fountain

Status: partial - ?

When

Scaffold / experimental stub stub/fountain (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/fountain"
```

```
cd 49-stub-fountain && source ../env.sh && go run .
```

Why

Recover specialist weight blobs over lossy links via LT fountain codes, then ensemble.

What

NewLTEncoder/Decoder, RecoverWeightBlobs, Neural/DenseFactory helpers.

Sample output (captured)

```
*fountain.LTEncoder <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/49-stub-fountain).

Live output

```
*fountain.LTEncoder <nil>
```

50. stub/hardware

suite=velvet slug=50-stub-hardware status=PASS time=0.05s

README

50. stub/hardware

Part: VIII - Stubs

Package: github.com/openfluke/welvet/stub/hardware

Status: partial - ?

When

Scaffold / experimental stub stub/hardware (may be partial).

Where

import "github.com/openfluke/welvet/stub/hardware"

```
cd 50-stub-hardware && source ../env.sh && go run .
```

Why

Dispatchers and UIs need a portable host audit (OS/CPU/RAM/GPU).

What

Audit() -> SystemAudit with linux /proc or platform fallbacks.

Sample output (captured)

```
{Model: Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz Logical:12 GOMAXPROCS:12}
OS: linux | CPU: Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz (12) | RAM: 31.12 GB | GPU: Unknown GPU (0.0 GB)
```

Live capture from go run ./cmd/runall on local ../../welvet (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/50-stub-hardware).

Live output

```
{Model: Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz Logical:12 GOMAXPROCS:12}
OS: linux | CPU: Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz (12) | RAM: 31.12 GB | GPU: Unknown GPU (0.0 GB)
```

51. stub/accel — NPU/Metal/QNN

suite=velvet slug=51-stub-accel status=PASS time=0.04s

README

51. stub/accel - NPU/Metal/QNN

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/accel

Status: missing -

When

Scaffold / experimental stub stub/accel (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/accel"
```

```
cd 51-stub-accel && source ../env.sh && go run .
```

Why

Vendor accelerators (Intel NPU, Qualcomm QNN, Apple Metal) will plug beside WebGPU - not replace it.

What

Package is doc.go only today (). No symbols yet; reserved for plugin loader design.

_No captured output in _manifest.json yet - run go run . and paste here._

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/51-stub-accel).

Sample output (captured)

```
- 51-stub-accel (83ms)
```

Live capture from go run ./cmd/runall on local ../velvet (exit 0).

Live output

(no output)

52. stub/clustering

suite=velvet slug=52-stub-clustering status=PASS time=0.04s

README

52. stub/clustering

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/clustering

Status: partial - ?

When

Scaffold / experimental stub stub/clustering (may be partial).

Where

import "github.com/openfluke/velvet/stub/clustering"

```
cd 52-stub-clustering && source ../env.sh && go run .
```

Why

Offline clustering helpers on tensors without inventing a second math stack.

What

KMeansCluster, HierarchicalGroup, distance/silhouette helpers.

Sample output (captured)

```
2 [0 1]
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/52-stub-clustering).

Live output

```
2 [1 0]
```

53. stub/ensemble

suite=welvet slug=53-stub-ensemble status=PASS time=0.04s

README

53. stub/ensemble

Part: VIII - Stubs

Package: github.com/openfluke/welvet/stub/ensemble

Status: partial - ?

When

Scaffold / experimental stub stub/ensemble (may be partial).

Where

```
import "github.com/openfluke/welvet/stub/ensemble"
```

```
cd 53-stub-ensemble && source ../env.sh && go run .
```

Why

Combine multiple model votes and find complementary specialists.

What

MajorityVote, FindComplementaryMatches, PerformanceSimilarity.

Sample output (captured)

```
[0 1 1]
```

Live capture from go run ./cmd/runall on local ../../welvet (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/53-stub-ensemble).

Live output

```
[0 1 1]
```

54. stub/evaluation

suite=velvet slug=54-stub-evaluation status=PASS time=0.05s

README

54. stub/evaluation

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/evaluation

Status: partial - ?

When

Scaffold / experimental stub stub/evaluation (may be partial).

Where

import "github.com/openfluke/velvet/stub/evaluation"

```
cd 54-stub-evaluation && source ../env.sh && go run .
```

Why

Benchmark grids through runtime/forward with deviation metrics.

What

EvaluateNetwork, MultiNetworkEvaluation, DeviationMetrics.

Sample output (captured)

```
<nil> evaluation: length mismatch inputs=1 expected=4
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/54-stub-evaluation).

Live output

```
<nil> evaluation: length mismatch inputs=1 expected=4
```

55. stub/grafting

suite=velvet slug=55-stub-grafting status=PASS time=0.05s

README

55. stub/grafting

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/grafting

Status: partial - ?

When

Scaffold / experimental stub stub/grafting (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/grafting"
```

```
cd 55-stub-grafting && source ../env.sh && go run .
```

Why

Merge grids into Parallel/Residual structures for topology experiments.

What

GraftGrids, ResidualGraft, GraftToGrid - Dense branches in v0.

Sample output (captured)

```
false grafting: no valid branches
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/55-stub-grafting).

Live output

```
false grafting: no valid branches
```

56. stub/grouping

suite=velvet slug=56-stub-grouping status=PASS time=0.04s

README

56. stub/grouping

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/grouping

Status: partial - ?

When

Scaffold / experimental stub stub/grouping (may be partial).

Where

import "github.com/openfluke/velvet/stub/grouping"

```
cd 56-stub-grouping && source ../env.sh && go run .
```

Why

Detect layer archetypes from safetensor-style names before mounting.

What

GroupRelatedTensors, DetectMHA/DetectSwiGLU/DetectRMSNorm -> ArchetypeHint.

Sample output (captured)

```
true {MultiHeadAttention 64 4}
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/56-stub-grouping).

Live output

```
true {MultiHeadAttention 64 4}
```

57. stub/introspection

suite=velvet slug=57-stub-introspection status=PASS time=0.04s

README

57. stub/introspection

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/introspection

Status: partial - ?

When

Scaffold / experimental stub stub/introspection (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/introspection"
```

```
cd 57-stub-introspection && source ../env.sh && go run .
```

Why

UIs and FFI need to list Grid methods without hardcoding every export.

What

GetMethods, GetMethodsJSON, GetMethodSignature.

Sample output (captured)

```
9 <nil>
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/57-stub-introspection).

Live output

```
9 <nil>
```

58. stub/observer

suite=velvet slug=58-stub-observer status=PASS time=0.06s

README

58. stub/observer

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/observer

Status: partial - ?

When

Scaffold / experimental stub stub/observer (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/observer"
```

```
cd 58-stub-observer && source ../env.sh && go run .
```

Why

Attach forward/backward observers for debugging without coupling to tanhi UDP.

What

Observer interface; ConsoleObserver / HTTPObserver / BufferObserver; ComputeLayerStats.

Sample output (captured)

```
*observer.ConsoleObserver
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/58-stub-observer).

Live output

```
*observer.ConsoleObserver
```

59. stub/pipeline

suite=velvet slug=59-stub-pipeline status=PASS time=0.04s

README

59. stub/pipeline

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/pipeline

Status: partial - ?

When

Scaffold / experimental stub stub/pipeline (may be partial).

Where

```
import "github.com/openfluke/velvet/stub/pipeline"
```

```
cd 59-stub-pipeline && source ../env.sh && go run .
```

Why

Decoder wavefront stats helpers - not a full Lucy-style pipeline runner yet.

What

PipelineForwardStats, TokenTimelineSummary.

Sample output (captured)

```
{PipelineTicks:0 SubLayerOps:0 MaxActiveJobs:0 MaxBlockSpread:0 MaxDistinctBlocks:0 MaxPendingTokens:0 StallFallback:false TokenDoneTick:[]}
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/59-stub-pipeline).

Live output

```
{PipelineTicks:0 SubLayerOps:0 MaxActiveJobs:0 MaxBlockSpread:0 MaxDistinctBlocks:0 MaxPendingTokens:0 StallFallback:false TokenDoneTick:[]}
```

60. stub/templates

suite=velvet slug=60-stub-templates status=PASS time=0.04s

README

60. stub/templates

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/templates

Status: partial - ?

When

Scaffold / experimental stub stub/templates (may be partial).

Where

import "github.com/openfluke/velvet/stub/templates"

```
cd 60-stub-templates && source ../env.sh && go run .
```

Why

Chat prompts must match model families (ChatML, Llama3, BitNet) without app-specific string glue.

What

Template.BuildPrompt; presets ChatML, Llama3, BitNetInstruction, MicrosoftBitNetChat.

Sample output (captured)

```
<|im_start|>system
You are helpful.
<|im_start|>user
Say hi
<|im_start|>assistant
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/60-stub-templates).

Live output

```
<|im_start|>system
You are helpful.
<|im_start|>user
Say hi
<|im_start|>assistant
```

61. stub/universal

suite=velvet slug=61-stub-universal status=PASS time=0.05s

README

61. stub/universal

Part: VIII - Stubs

Package: github.com/openfluke/velvet/stub/universal

Status: partial - ?

When

Scaffold / experimental stub stub/universal (may be partial).

Where

import "github.com/openfluke/velvet/stub/universal"

```
cd 61-stub-universal && source ../env.sh && go run .
```

Why

Probe unknown safetensor geometry and mount placeholder grids until full weight import lands.

What

LoadUniversal / LoadUniversalDetailed, ProbeDeepGeometry, MountGeometrically.

Sample output (captured)

```
false universal: "/path/to/snapshot": open /path/to/snapshot: no such file or directory
```

Live capture from go run ./cmd/runall on local ../../velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/61-stub-universal).

Live output

```
false universal: "/path/to/snapshot": open /path/to/snapshot: no such file or directory
```

62. w2a — validation harness

suite=velvet slug=62-w2a status=PASS time=0.04s

README

62. w2a - validation harness

Part: IX - Validate

Package: github.com/openfluke/w2a

Status: ok - [OK] harness

When

You need github.com/openfluke/w2a.

Where

```
import "github.com/openfluke/welvet/github.com/openfluke/w2a"
```

```
cd 62-w2a && source ../env.sh && go run .
```

Why

Engine packages must stay free of tests. w2a owns timed $34 \times 20 \times 3$ matrices, gap census, honesty stamps, and the train-mode permutation smoke (Test49). See §63 for a live full-suite run.

What

Interactive `go run .` ([0] Run ALL). Suites under `suites/*`. `StampBackendNote` / `AffinePackable` prevent fake [OK]. Test49: `AllNamedTrainModes` $\times 1^3/2^3/3^3 \times \text{Parallel/Bicameral/poly}$, origin-only - included in [0].

Sample output (captured)

```
w2a is a separate module - engine packages never contain tests.
cd w2a
go run .                                # interactive; [0] = ALL
go test ./tests/dense -v
go test ./tests/parallel -run Test49AllTrainModesCubes -count=1 -v
```

Live capture from `go run ./cmd/runall` on local `../..../velvet` (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/welvet/examples/62-w2a).

Live output

```
w2a is a separate module - engine packages never contain tests.
cd w2a
go run .                                # interactive; [0] = ALL
go test ./tests/dense -v
go test ./tests/parallel -run Test49AllTrainModesCubes -count=1 -v
```

63. Validation report — full suite

suite=welvet slug=63-validation status=PASS time=0.04s

README

63. Validation report - full suite

Part: IX - Validate

Package: github.com/openfluke/w2a

Status: ok - [OK] 246k cells

When

You need github.com/openfluke/w2a.

Where

```
import "github.com/openfluke/welvet/github.com/openfluke/w2a"
```

```
cd 63-validation && source ../env.sh && go run .
```

Why

Claims are cheap; a stamped matrix is not. This is the actual output of one full w2a [0] Run ALL so the book's [OK] marks are backed by numbers you can reproduce, not asserted.

What

Every timed layer sweeps its dtype × format × backend matrix; every suite runs its case checks. v1.0 board: 246,032 matrix cells, FAIL 0, RESULT PASS. GAP cells are declared skips (not silent fails).

Sample output (captured)

```
w2a [0] ALL: 246032 cells  FAIL 0  RESULT PASS
GAP = declared skip (GDN non-f32, AffinePacked, ...) - not a fail
```

Live capture from `go run ./cmd/runall` on local `../..welvet` (exit 0).

Source

Copied from the Welvet feature book examples (openfluke.github.io/welvet/examples/63-validation).

Live output

```
w2a [0] ALL: 246032 cells  FAIL 0  RESULT PASS
GAP = declared skip (GDN non-f32, AffinePacked, ...) - not a fail
```

64. Scorecard → v1.0 / minors

suite=velvet slug=64-scorecard status=PASS time=0.04s

README

64. Scorecard -> v1.0 / minors

Part: IX - Validate

Package: -

Status: ok - v1.1.0

When

Orientation chapter: Scorecard -> v1.0 / minors.

Where

Book / repo map (no single import)

```
cd 64-scorecard && source ../env.sh && go run .
```

Why

Version is earned from a weighted board, not marketing. v1.0 is 100/100 on the engine board. Minor tags (v1.1.0, ...) pack features without a new board. Apps, stubs, and NPU sit off-board.

What

version = 0.{round(earned)} until 100 -> v1.0. Scorecard today 100/100; this book tags v1.1.0. Training credit is \$9 on the board (8 pts), not an afterthought.

Sample output (captured)

```
v1.1.0
scorecard 100/100
- 64-scorecard (122ms)
===== velvet book: ok=70 fail=0 skip=0 =====
```

Live capture from go run ./cmd/runall on local .././velvet (exit 0).

Source

Copied from the Velvet feature book examples (openfluke.github.io/velvet/examples/64-scorecard).

Live output

```
v1.1.0
scorecard 100/100
```

71. runtime/dispatch — Op switchboard

suite=velvet slug=71-dispatch status=PASS time=0.05s

README

71. runtime/dispatch - Op switchboard

When: walking a grid / Parallel / Sequential and you only have any cells.

Where: github.com/openfluke/velvet/runtime/dispatch

Why: one Forward/Backward/Pack/SetDType/ApplyGrad path for every Velvet Op; unknown types hard-error (never silently pretend they are Dense).

Do not import dispatch from layers/parallel (import cycle) - Parallel dispatches branches itself.

```
cd 71-dispatch && source ../env.sh && go run .
```

Live output

```
dispatch dense [1 4] gIn 4 gW 16
dispatch cnn2 [1 2 2 2]
dispatch rejects unknown: dispatch: unsupported Op string (no silent Dense fallback)
```

72. model/wav2vec2 — CTC ASR

suite=welvet slug=72-wav2vec2 status=PASS time=0.05s

README

72. model/wav2vec2 - CTC ASR

When: speech to text (greedy CTC) from 16 kHz mono WAV/PCM.

Where: github.com/openfluke/welvet/model/wav2vec2

Why: run facebook/wav2vec2-base-960h in pure Go.

Always proves Config + Vocab + LoadHFDir error path. Full ASR needs HF snapshot via WAV2VEC2_DIR / WAV2VEC2_WAV.

```
cd 72-wav2vec2 && source ../env.sh && go run .
```

Live output

```
Base960h hidden=768 layers=12 heads=12 vocab=32
```

```
DecodeCTCGreedy: ATE
```

```
LoadHFDir without weights: open /home/openfluke/git/example/welvet/.cache/tmp/welvet-no-wav2vec2-weights/config.json: no such file or directory
```

```
set WAV2VEC2_DIR (+ WAV2VEC2_WAV) for full ASR smoke
```

73. All layers — one forward each

suite=welvet slug=73-all-layers status=PASS time=0.05s

README

73. All layers - one forward each

When: you want a single command that proves every layers/* Op (plus seqmix + dispatch) actually forwards.

Where: this folder imports every concrete layer package.

Why: chapters 11-28 split by topic; this is the regression checklist for "does it run?"

Cameral proofs (Freeze / CamSync on each Op) live in ../../cam/05_layers.

```
cd 73-all-layers && source ../env.sh && go run .
```

Each line prints when / where / why for that Op.

Live output

OK	dense	when=linear / FFN / proj where=layers/dense why=MatVec microkernel everything else bui
OK	ldns on	
OK	cnn1	when=1D seq / audio patches where=layers/cnn1 why=im2col->Dense Proj
OK	cnn2	when=images / 2D maps where=layers/cnn2 why=vision stem
OK	cnn3	when=volumes / voxels where=layers/cnn3 why=3D conv
OK	convt1	when=1D upsample where=layers/convt1 why=decoder / generative 1D
OK	convt2	when=2D upsample where=layers/convt2 why=decoder / U-Net expand
OK	convt3	when=3D upsample where=layers/convt3 why=volumetric decoder
OK	mha	when=token mixing via attention where=layers/mha why=transformers
OK	swiglu	when=gated FFN where=layers/swiglu why=modern LLM MLP
OK	rmsnorm	when=stabilize activations where=layers/rmsnorm why=LLM-style norm
OK	layernorm	when=stabilize activations where=layers/layernorm why=classic transformer norm
OK	softmax	when=distributions / heads where=layers/softmax why=classify / combine
OK	embedding	when=token -> vector where=layers/embedding why=lookup tables
OK	rnn	when=short temporal where=layers/rnn why=classic recurrent
OK	lstm	when=gated temporal where=layers/lstm why=longer memory than RNN
OK	sequential	when=stack Dense depth where=layers/sequential why=deep hemisphere body
OK	residual	when=skip + body where=layers/residual why=stable depth
OK	parallel	when=multi-cam / MoE where=layers/parallel why=combine cams + CamSync host
OK	kmeans	when=prototypes / codebook where=layers/kmeans why=cluster features
OK	mamba	when=selective SSM where=layers/mamba why=linear-time seq mix
OK	gdn	when=gated delta / linear attn where=layers/gdn why=decode-friendly mixer
OK	metacognition	when=observer / confidence where=layers/metacognition why=meta signals beside main pat
OK	h	
OK	seqmix	when=name the mixer kind where=layers/seqmix why=contract only - swap MHA/Mamba/GDN
OK	dispatch	when=any Op -> Forward where=runtime/dispatch why=grid walk without type switches in y
OK	our code	
all-layers: ok=24 fail=0		

74. Kokoro — empty model vs apps

suite=velvet slug=74-kokoro status=PASS time=0.04s

README

74. Kokoro - reserved empty slots

When: you specifically want Kokoro TTS.
Where: velvet/model/kokoro and velvet/apps/kokoro - both empty placeholders today.
Why: names reserved for a future port; do not import them yet.

Working speech apps right now:

App	Role
`apps/mosstts`	Moss TTS shell
`apps/qwentts`	Qwen TTS shell

```
cd 74-kokoro && source ../env.sh && go run .
```

Live output

When: you want Kokoro-style TTS
Where: planned at model/kokoro + apps/kokoro - both empty today
Why: placeholders reserved; working TTS apps are mosstts / qwentts
model/kokoro entries: 0
apps/kokoro entries: 0
OK working TTS shell: apps/mosstts
OK working TTS shell: apps/qwentts
OK - kokoro slots empty; use mosstts/qwentts until kokoro lands

75. Apps map — when/where/why

suite=velvet slug=75-apps-map status=PASS time=0.04s

README

75. Apps map - when / where / why

When: you need a product shell (chat, TTS, ASR, image) rather than a library Op.

Where: velvet/apps/<name> - each is its own Go module.

Why: engine packages never import apps; apps replace -> velvet.

This chapter asserts every known app directory exists and prints guidance.

It does not download model weights or start interactive CLIs.

```
cd 75-apps-map && source ../env.sh && go run .
```

Live output

Apps never import into the engine - run each from its own go.mod.

```
OK  octo           [go.mod ok]
    when: chat / agent CLI over transformer entities
    where: apps/octo
    why:  main interactive shell
OK  flux2          [no go.mod]
    when: image generation experiments
    where: apps/flux2
    why:  diffusion-style app shell
OK  mosstts        [no go.mod]
    when: speech synthesis demos
    where: apps/mosstts
    why:  TTS product path
OK  kokoro         [empty placeholder]
    when: Kokoro TTS (placeholder - empty dir)
    where: apps/kokoro
    why:  reserved; use mosstts/qwentts until filled
OK  aai            [no go.mod]
    when: AAI / Lucy organism demos
    where: apps/aai
    why:  cameral + lucy product
OK  biotalk        [go.mod ok]
    when: bio / organism chat surfaces
    where: apps/biotalk
    why:  domain UI over velvet
OK  qwenasr        [no go.mod]
    when: Qwen ASR
    where: apps/qwenasr
    why:  speech recognition app
OK  qwentts        [no go.mod]
    when: Qwen TTS
    where: apps/qwentts
    why:  speech synthesis app
OK  planetbridging [go.mod ok]
    when: planetbridging demos
    where: apps/planetbridging
    why:  bridging experiments
```

All app folders present. Deep cameral proofs: example/cam/

01. Train modes

suite=cam slug=01_modes status=PASS time=8.47s

README

01 - BranchModes

When: different update rules (or idle/teacher) per cam

Where: para.SetBranchModes(...) with parent NormalBP

Why: mix credit, freeze priors, distill, adversarial twins, surprise memory

```
go run ./01_modes
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 01_modes - prove BranchModes ===
PASS both_BP both cams moved  $\Delta=0.009232$  /  $0.009232$ 
PASS Freeze cam0 moved  $\Delta=0.009794$ , frozen cam1  $\Delta=0$  (want  $\sim 0$ )
PASS Shadow teacher frozen  $\Delta=0$ , student moved  $\Delta=0.1343$ 
PASS Adversarial both move  $\Delta=0.00832/0.00832$ ; Adv loss  $0.0013 \geq$  BP loss  $0.0007$  (fight)
PASS Memory asleep  $\Delta=0$  ( $\sim 0$ ), awake  $\Delta=0.007141$  ( $>0$ )
PASS Tween Tween cam moved  $\Delta=0.003871$ 
all BranchMode proofs passed
```

Why these PASS lines prove it

Proof | Assertion | Why that shows the feature

****both_BP**** | both cams $\Delta W \approx 0$ | Shared learning works

****Freeze**** | cam0 moves, cam1 $\Delta \approx 0$ | Frozen cam still forwards; no weight update

****Shadow**** | teacher $\Delta \approx 0$, student moves | Shadow = frozen KD teacher

****Adversarial**** | both move; Adv loss $\geq$ BP loss | Negated LR fights the objective

****Memory**** | asleep $\Delta \approx 0$, awake $\Delta \approx 0$ | SurpriseThresh gates updates

****Tween**** | Tween cam $\Delta \approx 0$ | Alternate update family still applies

Cams are diverged first so Freeze isn't vacuous.

Live output

```
=== 01_modes - prove BranchModes ===
PASS both_BP both cams moved  $\Delta=0.009232$  /  $0.009232$ 
PASS Freeze cam0 moved  $\Delta=0.009794$ , frozen cam1  $\Delta=0$  (want  $\sim 0$ )
PASS Shadow teacher frozen  $\Delta=0$ , student moved  $\Delta=0.1343$ 
PASS Adversarial both move  $\Delta=0.00832/0.00832$ ; Adv loss  $0.0013 \geq$  BP loss  $0.0007$  (fight)
PASS Memory asleep  $\Delta=0$  ( $\sim 0$ ), awake  $\Delta=0.007141$  ( $>0$ )
PASS Tween Tween cam moved  $\Delta=0.003871$ 
all BranchMode proofs passed
```

02. Combine strategies

suite=cam slug=02_combine status=PASS time=1.24s

README

02 - Combine modes

When: choosing how hemisphere posts become one tensor

Where: Config.Combine

Why: avg/add ensemble; max WTA; sparsek; disagree; filter MoE; concat widths

```
go run ./02_combine
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 02_combine - prove merge modes ===
PASS avg_width outShape=[1 4] want [1 4]
PASS add_width outShape=[1 4] want [1 4]
PASS concat_width outShape=[1 8] want feat=8
PASS max_routes_to_cam0 combined should match cam0 when cam0cam1
PASS sparsek_keeps_strong SparseK=1 should ≈ strongest cam (cam0)
PASS disagree_formula y = mean + β(a-b) holds
PASS filter_gate_trains MoE gate Δ=0.01582
all Combine proofs passed
```

Why these PASS lines prove it

Proof | Assertion | Why

avg/add_width | out feat = 4 | Same-width merge

concat_width | out feat = 8 | Unequal stack = 4+4

max_routes_to_cam0 | combined $\equiv$ cam0 when cam0cam1 | Hard max routing

sparsek_keeps_strong | SparseK=1 $\approx$ strongest cam | Top-K by $\|out\|_2$

disagree_formula | $y = \text{mean} + \beta(a-b)$ numerically | Debate combine is exact

filter_gate_trains | MoE gate weights move | Gate is live, not decoration

Live output

```
=== 02_combine - prove merge modes ===
PASS avg_width outShape=[1 4] want [1 4]
PASS add_width outShape=[1 4] want [1 4]
PASS concat_width outShape=[1 8] want feat=8
PASS max_routes_to_cam0 combined should match cam0 when cam0cam1
PASS sparsek_keeps_strong SparseK=1 should ≈ strongest cam (cam0)
PASS disagree_formula y = mean + β(a-b) holds
PASS filter_gate_trains MoE gate Δ=0.01582
all Combine proofs passed
```

03. CamSync

suite=cam slug=03_camsync status=PASS time=1.51s

README

03 - CamSync

When: cams should share / pull weight DNA

Where: SetCamSync / SyncNow / Cross

Why: soft ↔ hard consensus, one-way teacher, groups, cross-layer

```
go run ./03_camsync
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 03_camsync - prove sync pulls cos up ===
PASS  α=1_hard_sync   cos 0.3769 -> 1.0000 (hard sync should ≈1 and beat pre)
PASS  α=0.25_soft_pull cos 0.3717->0.6413, still distinct ΔW=0.75
PASS  BranchAlpha_one_way  cam1 unchanged Δ=0; cos 0.3717->0.6635 (cam0 pulled to mean)
PASS  Groups_exclude_cam2  cam2 frozen Δ=0; cam0≡cam1 Δ=0; cam0≠cam2 Δ=3 (was 3)
PASS  Cross_pair   Cross hard-sync cos 0.0000->1.0000
PASS  Cross_stack_trains  TrainStackMSE err=<nil>
all CamSync proofs passed
```

Why these PASS lines prove it

Proof | Assertion | Why

```
**α=1_hard_sync** | cos 0.37 -> **1.0** | Diverged cams hard-average to identical
**α=0.25_soft_pull** | cos rises, weights still distinct | Soft pull ≠ clone
**BranchAlpha_one_way** | cam1 Δ=0, cos rises | Teacher not overwritten; student pulls to mean
**Groups_exclude_cam2** | cam0≡cam1, cam2 frozen & different | Clique sync only
**Cross_pair** | Cross hard-sync cos -> 1 | Stack endpoint glue works
**Cross_stack_trains** | TrainStackMSE ok | Training still works with Cross cfg
```

Live output

```
=== 03_camsync - prove sync pulls cos up ===
PASS  α=1_hard_sync   cos 0.3769 -> 1.0000 (hard sync should ≈1 and beat pre)
PASS  α=0.25_soft_pull cos 0.3717->0.6413, still distinct ΔW=0.75
PASS  BranchAlpha_one_way  cam1 unchanged Δ=0; cos 0.3717->0.6635 (cam0 pulled to mean)
PASS  Groups_exclude_cam2  cam2 frozen Δ=0; cam0≡cam1 Δ=0; cam0≠cam2 Δ=3 (was 3)
PASS  Cross_pair   Cross hard-sync cos 0.0000->1.0000
PASS  Cross_stack_trains  TrainStackMSE err=<nil>
all CamSync proofs passed
```

04. CamKit

suite=cam slug=04_kit status=PASS time=1.34s

README

04 - CamKit / LRs / Rotate / DNA / Dream

When: schedules, plasticity, replay, speciation

Where: SetBranchLRs / SetRotateSchedule / SetCamKit / DreamPulse

Why: sleep cycles, soft freeze, DNA push/pull, offline consolidation

```
go run ./04_kit
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 04_kit - prove LRs / rotate / DNA / dream ===
PASS BranchLRs_1|0 cam0 Δ=0.009794 cam1 Δ=0
PASS Rotate_slot0 first 5: cam0 Δ=0.002567 cam1 Δ=0
PASS Rotate_slot1 next 5: cam0 Δ=0 cam1 Δ=0.002486
PASS DNAREg_diversify diversify cos 0.8446 -> -0.8649
PASS DNAREg_attract attract cos 0.3717 -> 0.9994
PASS DreamPulse replay avgLoss=0.0012 moved cam0 Δ=0.0009936
PASS RefreshMetrics modes=[NormalBP Freeze] plast=[0.000660944211525738 0]
all CamKit proofs passed
```

Why these PASS lines prove it

Proof | Assertion | Why

****BranchLRs 1\ | 0**** | cam1 Δ=0 | LR×0 = soft freeze

****Rotate_slot0/1**** | only cam0 then only cam1 moves | Sleep schedule flips plasticity

****DNAREg_diversify**** | cos 0.84 -> ****negative**** | Push away from mean

****DNAREg_attract**** | cos 0.37 -> ****~1**** | Pull toward mean

****DreamPulse**** | replay loss>0 and ΔW>0 | Buffer replay moves weights

****RefreshMetrics**** | modes show Freeze, plast[1]=0 | Meter matches reality

Live output

```
=== 04_kit - prove LRs / rotate / DNA / dream ===
PASS BranchLRs_1|0 cam0 Δ=0.009794 cam1 Δ=0
PASS Rotate_slot0 first 5: cam0 Δ=0.002567 cam1 Δ=0
PASS Rotate_slot1 next 5: cam0 Δ=0 cam1 Δ=0.002486
PASS DNAREg_diversify diversify cos 0.8446 -> -0.8649
PASS DNAREg_attract attract cos 0.3717 -> 0.9994
PASS DreamPulse replay avgLoss=0.0012 moved cam0 Δ=0.0009936
PASS RefreshMetrics modes=[NormalBP Freeze] plast=[0.000660944211525738 0]
all CamKit proofs passed
```

05. Layers as cams

suite=cam slug=05_layers status=PASS time=1.48s

README

05 - Every layer as cams

When: CNN / Mamba / LSTM / ... as Parallel cams

Where: HemispheresFrom + twin Ops

Why: CamSync / Freeze / Train work on every hosted Op

```
go run ./05_layers
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 05_layers - prove each Op works as cams ===
PASS dense default host loss 0.1230->0.0789 Freeze ok
PASS cnn1 1D conv loss 0.1181->0.0854 Freeze ok
PASS cnn2 2D vision loss 0.1161->0.0602 Freeze ok
PASS cnn3 3D loss 0.1155->0.0271 Freeze ok
PASS convt1 1D upsample loss 0.1206->0.1148 Freeze ok
PASS convt2 2D upsample loss 0.1201->0.1077 Freeze ok
PASS convt3 3D upsample loss 0.1188->0.1053 Freeze ok
PASS mha attention loss 0.1217->0.1195 Freeze ok
PASS swiglu FFN loss 0.1230->0.1223 Freeze ok
PASS rmsnorm norm loss 0.4835->0.0027 Freeze ok
PASS layernorm norm loss 1.1225->0.0003 Freeze ok
PASS softmax distribution loss 0.0708->0.0708 Freeze ok
PASS rnn temporal loss 0.1225->0.0009 Freeze ok
PASS lstm gated temporal loss 0.1225->0.0741 Freeze ok
PASS embedding tables loss 0.1252->0.0100 Freeze ok
PASS sequential deep hemi loss 0.1218->0.1203 Freeze ok
PASS residual residual hemi loss 0.0431->0.0277 Freeze ok
PASS kmeans prototypes loss 0.0003->0.0003 Freeze ok
PASS mamba SSM loss 0.1208->0.0996 Freeze ok
PASS metacognition observer loss 0.1225->0.0282 Freeze ok
PASS gdn gated delta loss 0.1225->0.1225 Freeze ok
all layer-as-cam proofs passed
```

Why these PASS lines prove it

Each layer must:

1. Forward without error
2. Learn (loss drops after seeded init) - except weightless softmax, tiny kmeans, zero-blob gdn (Forward+Freeze still required)
3. Freeze cam1 - ActiveModes[1]==Freeze and primary store $\Delta \approx 0$

PASS ... loss a->b Freeze ok means all three held. CNN/MHA/etc. are seeded (Velvet New is zero-init).

Live output

```
=== 05_layers - prove each Op works as cams ===
PASS dense default host loss 0.1230->0.0789 Freeze ok
PASS cnn1 1D conv loss 0.1181->0.0854 Freeze ok
PASS cnn2 2D vision loss 0.1161->0.0602 Freeze ok
PASS cnn3 3D loss 0.1155->0.0271 Freeze ok
```

```
PASS convt1 1D upsample loss 0.1206->0.1148 Freeze ok
PASS convt2 2D upsample loss 0.1201->0.1077 Freeze ok
PASS convt3 3D upsample loss 0.1188->0.1053 Freeze ok
PASS mha attention loss 0.1217->0.1195 Freeze ok
PASS swiglu FFN loss 0.1230->0.1223 Freeze ok
PASS rmsnorm norm loss 0.4835->0.0027 Freeze ok
PASS layernorm norm loss 1.1225->0.0003 Freeze ok
PASS softmax distribution loss 0.0708->0.0708 Freeze ok
PASS rnn temporal loss 0.1225->0.0009 Freeze ok
PASS lstm gated temporal loss 0.1225->0.0741 Freeze ok
PASS embedding tables loss 0.1252->0.0100 Freeze ok
PASS sequential deep hemi loss 0.1218->0.1203 Freeze ok
PASS residual residual hemi loss 0.0431->0.0277 Freeze ok
PASS kmeans prototypes loss 0.0003->0.0003 Freeze ok
PASS mamba SSM loss 0.1208->0.0996 Freeze ok
PASS metacognition observer loss 0.1225->0.0282 Freeze ok
PASS gdn gated delta loss 0.1225->0.1225 Freeze ok
all layer-as-cam proofs passed
```

06. Recipes

suite=cam slug=06_recipes status=PASS time=1.10s

README

06 - Recipes

When: compose modes + sync + kit into behaviors

Where: go run ./06_recipes

Why: teacher, sleep, debate, memory, dream, concat

```
go run ./06_recipes
```

Exits non-zero if any proof fails (PASS / FAIL lines).

Live output

```
=== 06_recipes - prove composed behaviors ===
PASS teacher_student teacher frozen, student moved
PASS sleep_slot0 only cam0 Δ=0.00257/0
PASS sleep_slot1 only cam1 Δ=0/0.00249
PASS debate both move; debate loss 0.0121 vs coop 0.0110
PASS surprise_memory asleep Δ=0 awake Δ=0.00582
PASS dream_consolidate avg=0.0012 weights moved on replay
PASS cross_modal_concat feat=8 want 3+5=8
PASS cross_modal_trains cam0 moved under concat
all recipe proofs passed
```

Why these PASS lines prove it

Proof | Assertion | Why

```
**teacher_student** | teacher frozen, student moved | Shadow + one-way sync
**sleep_slot0/1** | alternating single-cam ΔW | Rotate schedule
**debate** | both move; debate loss ≥ coop | Adv + Disagree
**surprise_memory** | asleep vs awake ΔW | SurpriseThresh gate
**dream_consolidate** | replay moves weights | DreamBuffer
**cross_modal_concat** | feat=8=3+5; cam0 trains | Unequal cams need concat
```

Live output

```
=== 06_recipes - prove composed behaviors ===
PASS teacher_student teacher frozen, student moved
PASS sleep_slot0 only cam0 Δ=0.00257/0
PASS sleep_slot1 only cam1 Δ=0/0.00249
PASS debate both move; debate loss 0.0121 vs coop 0.0110
PASS surprise_memory asleep Δ=0 awake Δ=0.00582
PASS dream_consolidate avg=0.0012 weights moved on replay
PASS cross_modal_concat feat=8 want 3+5=8
PASS cross_modal_trains cam0 moved under concat
all recipe proofs passed
```